INCH-POUND
MIL-DTL-16377/33B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/33A(SH)
9 May 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, DETAIL LIGHTING FOR SIGHTGLASSES, 6 WATTS, 120 VOLTS, WATERTIGHT, SYMBOLS 136.1, 156, 136.1SSL, AND 156SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 4 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

Distribution Statement A. Approved for public release: distribution is unlimited.

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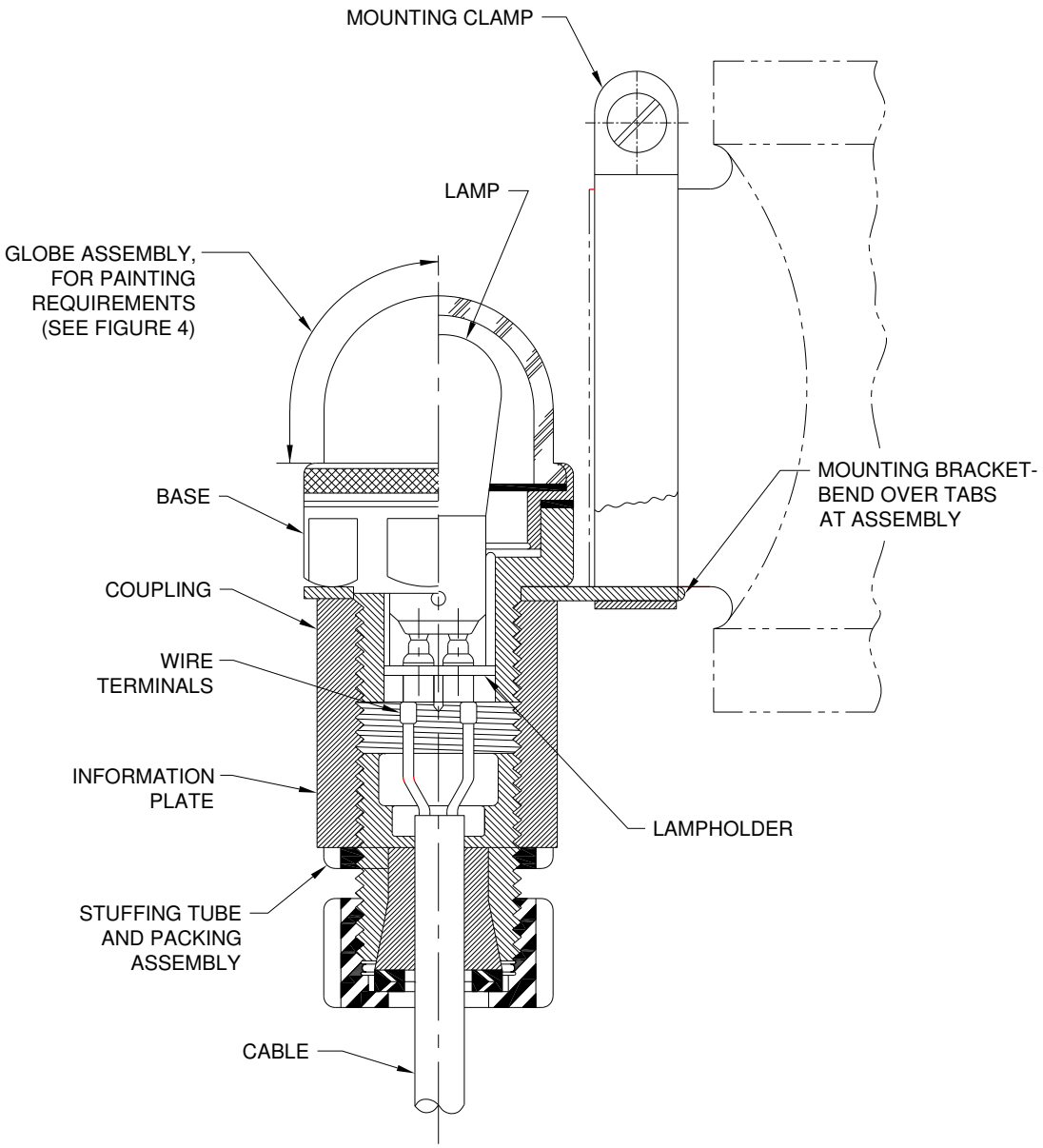


FIGURE 1. Typical view of symbols 136.1 and 156.

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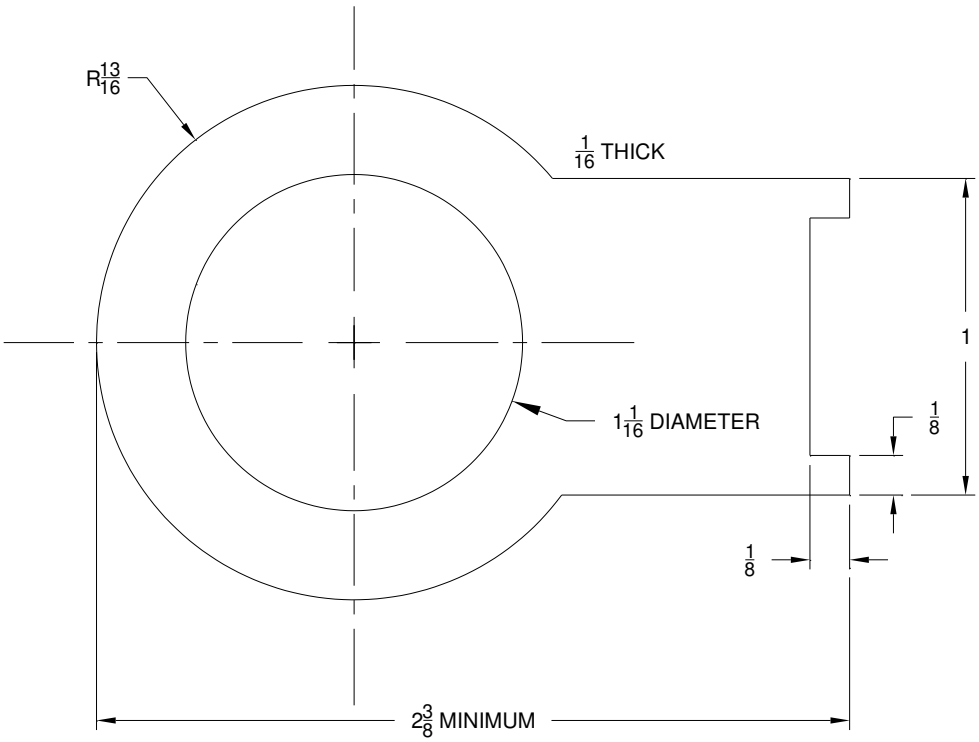


FIGURE 2. Mounting bracket.

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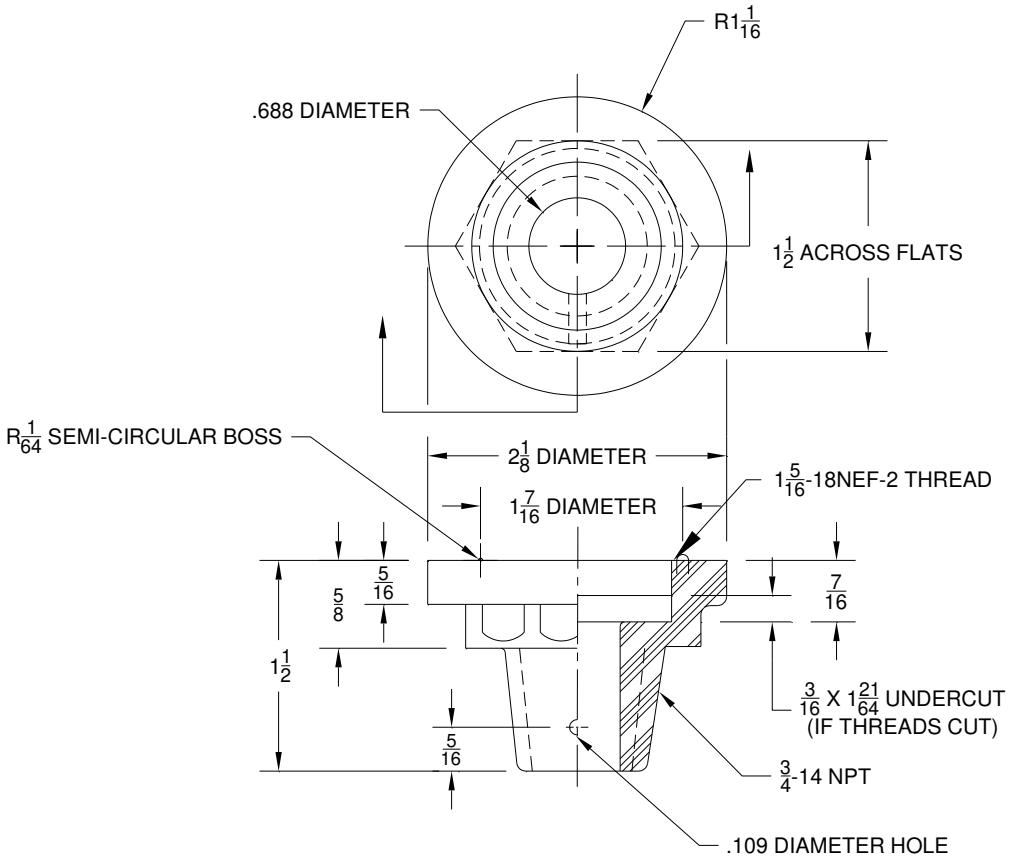
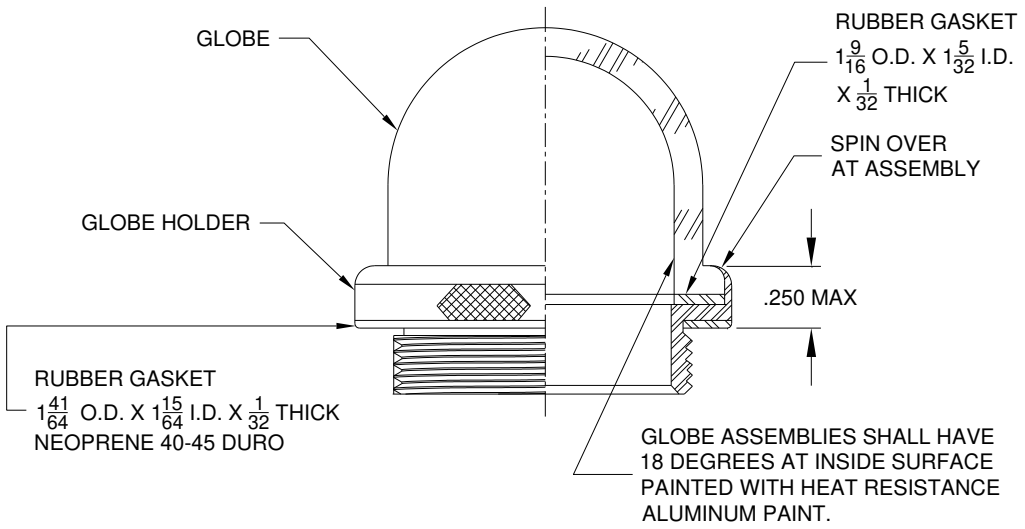
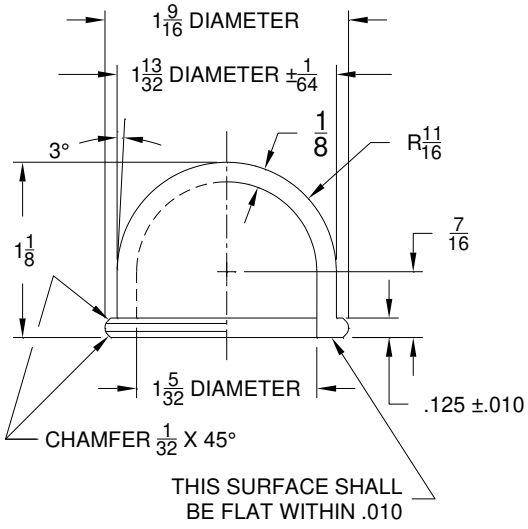


FIGURE 3. Base.

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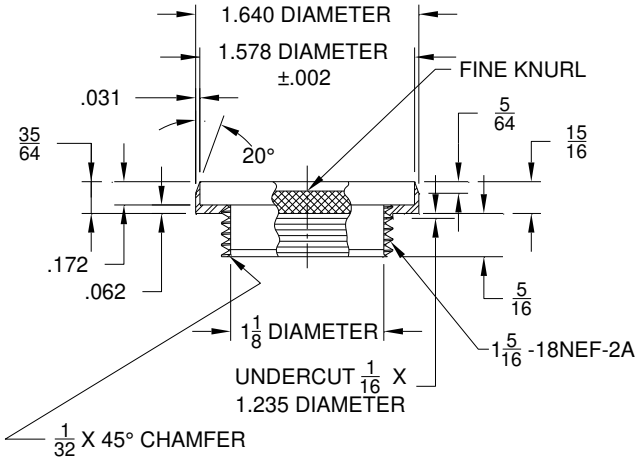


(a) GLOBE ASSEMBLY



MATERIAL - TEMPERED GLASS
COLOR:
FOR SYMBOL 136.1 - CLEAR TRANSPARENT
FOR SYMBOL 156 - RED TRANSPARENT
TRANSMITTANCE (SYMBOL 136.1):
CLEAR TRANSPARENT - 85 PERCENT MINIMUM

(b) GLOBE, GLASS



MATERIAL BASS

(c) GLOBE, HOLDER

FIGURE 4. Globe.

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INCANDESCENT REQUIREMENTS

CLASSIFICATION

Type: II
Class: 1

SYMBOL NUMBERS

136.1: Light for sightglasses, with clear transparent globe assembly, for use with one 6-watt, 120-volt white (clear) incandescent lamp.
156: Light for sightglasses, with red transparent globe assembly, for use with one 6-watt, 120-volt white (clear) incandescent lamp.

DESIGN

Electrical rating:	6 watts, 120 volts.
Dimensions and configuration:	See figures 1 through 4.
Mounting bracket:	Brass or corrosion-resistant steel. See figure 2.
Mounting clamp:	Steel, zinc plated, or corrosion-resistant steel, size 2¼ to 3¼ inside diameter.
Base:	Alkyd resin, color gray. See figure 3.
Coupling:	¾-inch iron pipe size, polyvinylchloride material.
Stuffing tube:	Size 2, ASTM F1836M
Packing assembly:	Size 2, ASTM F1836M
Globe assembly:	Symbol 136.1, clear transparent. See figure 4. Symbol 156, red transparent. See figure 4.
Lampholder:	UL 496, Supplement SB, bayonet candelabra, double-contact.
Cable:	10 feet of type LS2SWU, size 18 AWG, two-conductor, conforming to MIL-DTL-24643.
Wire terminals:	Underwriters Laboratories-approved insulated crimp type.
Lamp:	Lights shall be designed to accommodate a 6-watt, 120-volt, incandescent lamp conforming to industry number 6S6DC(120). (Lamp not furnished.)
Weight:	1 pound, 4 ounces maximum.
Information plate:	Brass, or corrosion-resistant steel, permanently mounted where shown on figure 1. Legend, in ⅛-inch characters, shall be as follows: “INSTALL 6-WATT LAMP ONLY: LAMP: NSN 6240-00-143-3060 MIL-DTL-16377/33 REV _____”

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 2

SYMBOL NUMBERS

136.1SSL: SSL for sightglasses, white.
156SSL: SSL for sightglasses, red.

DESIGN

Electrical rating:	6 watts maximum, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1 through 4.

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Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.6.17 of MIL-DTL-16377.
Mounting dimensions:	See figures 2 and 3.
Cable:	10 feet of type LS2SWU, size 18 AWG, two-conductor, conforming to MIL-DTL-24643.
Enclosure:	Watertight. See 3.5.11 of MIL-DTL-16377.
Weight:	1 pound, 4 ounces maximum.
Designation and marking:	One type I label.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I for type I and table III for type III.
Globes:	See table II.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection
Conformance inspection:	See MIL-DTL-16377.

GENERAL INFORMATION

Assignment of military part numbers and general information shall be as shown in table IV.

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TABLE I. First article and comparison inspection for complete type II fixture.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Color and luminous transmittance	3.7.3.1	4.8.6.3	---
Paint removal	3.7.5.5	4.8.17.2	---
Mechanical shock	3.7.5.6	4.8.17.3	6 foot-pounds
Thermal shock	3.7.5.7	4.8.17.4	---
Breakage pattern	3.7.5.9	4.8.17.6	---
Magnetic permeability	3.4.8	4.8.18	---
Continuity of grounding	3.6.13.4	4.8.12	---

TABLE II. First article and comparison inspection for globes.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.13	4.5	---
Optical uniformity	3.7.5.1	4.8.4	---
Color and luminous transmittance	3.7.3.1	4.8.6.3	---
Shock	3.5.12	4.8.7	---
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Paint removal	3.7.5.5	4.8.17.2	---
Mechanical shock	3.7.5.6	4.8.17.3	6 foot-pounds
Thermal shock	3.7.5.7	4.8.17.4	---
Breakage pattern	3.7.5.9	4.8.17.6	---

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TABLE III. First article and comparison inspection for complete type III fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Shock	3.5.12	4.8.7	Grade B 1/
Vibration	3.5.13	4.8.8	1/
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Continuity of grounding	3.6.13.4	4.8.12	1/
Photometrics	3.7.1	4.8.6	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature, white light only	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Optical uniformity	3.7.6.1.1.1	4.8.4	---
Stress relief	3.8.5	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

Military Part Number M16377/33-	Item Name	National Stock Number (NSN)
136.1	Complete fixture, symbol 136.1	6210-00-790-4846
156	Complete fixture, symbol 156	6210-00-790-4847
001	Globe assembly, clear	---
002	Globe assembly, red	
136.1SSL	Complete fixture, symbol 136.1SSL	---
156SSL	Complete fixture, symbol 156SSL	

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CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-013)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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